5-DOF Lamp Robot Character Simulation

Complete Setup, Deployment & Testing Guide for Ubuntu 24.04 Virtual Machines

System Summary: This project implements a fully offline, expressive 5-DOF lamp robot character. The architecture features a strict JSON protocol over a **TCP socket transport boundary**, an authoritative body receiver enforcing whole-command timeouts and enum limits, local AI models (Vosk for offline STT, Moondream for VLM vision, and Qwen2:0.5B for cognitive reasoning), cross-platform multi-modal audio/light synthesis, and a goal-directed observe-act-reobserve path.

1. Target Environment & Hardware Requirements

The system is optimized for a clean **Ubuntu 24.04 LTS** virtual machine (VMware, VirtualBox, KVM, or WSL2):

- **CPU:** 4 vCPU cores (x86_64)
- **Memory:** 8 GB RAM (runs comfortably without swap thrashing)
- **GPU:** No discrete GPU or CUDA required (all models run on CPU)
- **Peripherals:** Virtual or physical webcam, microphone, and audio speaker

2. Ubuntu System Prerequisites & Packages

Open a terminal on your Ubuntu machine and install the required build tools, audio, video, and TTS packages:

```
sudo apt update
sudo apt install -y python3-pip python3-dev portaudio19-dev espeak-ng ffmpeg alsa-utils unzip
wget curl git
```

3. Extracting Project & Installing Dependencies

Unzip the provided submission archive and install the Python dependencies:

```
unzip submission.zip -d lamp_robot
cd lamp_robot
pip3 install -r requirements.txt --break-system-packages
```

Note: On Ubuntu 24.04, Python 3.12+ enforces PEP 668. You can use a virtual environment or --break-system-packages.

4. Local Model Downloads (Vosk + Ollama)

A. Offline Audio Speech Recognition (Vosk)

Download and extract the lightweight English speech model into the model/ directory:

```
wget https://alphacephei.com/vosk/models/vosk-model-small-en-us-0.15.zip
unzip vosk-model-small-en-us-0.15.zip
mv vosk-model-small-en-us-0.15 model
rm vosk-model-small-en-us-0.15.zip
```

B. Local Vision & Cognition Models (Ollama)

Install Ollama and pull the required tiny vision and language models:

```
curl -fsSL https://ollama.com/install.sh | sh
# Verify Ollama is active
ollama pull moondream
ollama pull qwen2:0.5b
```

5. Running the Character Robot Simulation

Start the complete autonomous character loop with:

```
python3 main.py
```

Virtual Machine Environment Flags:

- FORCE_ENGAGEMENT=1: Bypasses webcam face detection and simulates active face engagement. Essential for headless servers, VMs without camera pass-through, or test scripts.
- PYBULLET_DIRECT=1: Runs PyBullet physics and kinematics in direct headless mode without creating an X11 window.

Example for testing in a VM without a physical camera:

```
FORCE_ENGAGEMENT=1 python3 main.py
```

6. Running the Automated Test Suite

The submission includes 21 comprehensive unit and integration tests across 5 test suites:

```
python3 -m unittest discover -s tests -v
```

Test Module	Coverage Area	Key Verifications
test_protocol.py	JSON Protocol & Schema	Command/result round-trip, enum validation, safe intent clamping, timeout limits.
test_transport.py	TCP Socket Boundary	Client/server message exchange over loopback TCP, timeout handling, rejection handling.
test_body_receiver.py	Authoritative Receiver	Sequence number checking, status tracking, whole-command timeout reporting, hardware safety.
test_action_controller.py	Multi-Modal Body Controller	PyBullet motion execution, light cues, PCM audio synthesis for Linux, TTS budget.
test_runtime_and_verification.py	Cognition & Verification	Target extraction, dynamic scene-aware search planning, completion verification.

7. Demonstration Moments & Raw Measurements

Each interaction records rich telemetry into measurements.jsonl. The system demonstrates 5 core moments:

1. **Engagement:** Face detection triggers awareness; disengages when user leaves.

2. **Character Response:** Intentional combination of motion (nod/search), warm/success lights, audio sound cues (chirp/confirm), and speech.
3. **Spoken Interaction:** Vosk offline STT transcribes microphone audio; TTS speaks responses.
4. **Scene Memory:** Moondream vision analyzes live camera frames; answers spoken questions referencing memory.
5. **Goal-Directed Action:** Evaluates goal target against live vision, searches scene dynamically if missing, and validates post-observation state.

8. Troubleshooting & Common VM Issues

Issue	Likely Cause	Solution
Connection refused on port 8765	Server starting or port conflict	Wait 1s for server bind or check port with lsof -i :8765.
Ollama model query fails	Ollama daemon not running	Run ollama serve or check systemctl status ollama.
Vosk STT skipped / silent	Missing 'model' folder	Ensure Vosk model is unpacked at ./model.
No audio sound in VM	VM audio device disabled	Enable Intel HD Audio or ALSA in VM hypervisor settings.